

Description of Classes

Sensor

The Sensor class is an abstraction which represents the interface with the third-party hardware which will do the actual animal detection. Each instance of the class will represent a single physical sensor and contain a representation of the most recent data received from that sensor. The mechanism by which data is received is yet to be determined, but one potential option is to have the sensor push updates via a remote procedure call. When the sensors report potential animal sounds of interest, they will be represented by Noise objects, a list of which will exist inside each Sensor object. If a noise with sufficient strength is detected, the sensor will create a Report object referring to it and place it in its outgoing box.

Noise

The Noise class represents individual potential animal sounds reported by the sensors. These are ephemeral and will be erased and recreated each time new data comes in.

Animal

The Animal class enumerates different types of animals which the system is capable of detecting, and exists mainly to avoid the use of magic numbers.

Report

The Report class represents a detection event of sufficient interest to trigger an alarm, and will be persisted as part of a FiledReport.

Alarm

The Alarm class is a singleton which is responsible for monitoring the various sensors for outgoing reports. This will be accomplished by polling each sensor object on a regular basis, perhaps once per second. If a new report is found it will be wrapped in a FiledReport object and added to the list in the Alerts object.

FiledReport

The FiledReport class represents past alarms. It stores a copy of the original report along with how it has been classified and the Ranger who classified it.

Alerts

The Alerts class is a singleton which is responsible for storing and retrieving reports. The mechanism by which data is persisted has yet to be determined, but one potential option is to use an object relational model and an SQL database.

Ranger

The Ranger class represents the users of the system, for access control and record keeping purposes.